

Isabelle / HOL Basics

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP)

1. Introduction & Functional Programming Concepts

Introducing Isabelle/HOL:

- KEATS / Course platform
- **Higher-Order Logic (HOL):**
 - Logic: $\forall, \exists, \vee, \wedge, \rightarrow, \dots$
 - Functional programming paradigm
- **Key Features:**
 - Define datatypes
 - Functions (recursive & non-recursive)
 - State & prove logical properties of functions

Functional Programming Concepts:

- **Currying:**
 - $\Rightarrow$ Maths textbook: $g(x, y), \quad h(x, y, z)$
 - $\Rightarrow$ Curried version: $g \ x \ y, \quad h \ x \ y \ z$
- **Lambda Expressions (λ):**
 - $\Rightarrow$ Maths textbook: $f(x, y) = 2x + y$
 - $\Rightarrow$ Lambda form: $f' = (\lambda x \ y. 2x + y)$
 - $\Rightarrow f(a, b) = 2a + b \iff (f' \ a \ b) = 2a + b$
 - $\Rightarrow$ Partial application: $f' \ a = (\lambda y. 2a + y)$

2. Types & Terms in Isabelle

Terms & Overloading:

- Terms include numbers, functions, lists, trees, etc.
- Booleans: `True`, `False`
- Numerals $0, 1, \dots$ and operators $(+), (-), (*)$ are **overloaded** across types (`nat`, `int`, `real`).
- *Note:* Blue terms in Isabelle PIDE indicate unbound / free variables.

Natural Numbers & Lists:

- **Naturals (`nat`):** either `"0"` or `"Suc a"` (where a is a natural number).
- **Lists (`'a list`):**
 - Empty: `[]`
 - Non-empty: `Cons x xs` (or `x # xs`)
 - $[a, b, c] \sim \text{Cons } a (\text{Cons } b (\text{Cons } c []))$

Type Annotation Syntax (τ):

- **Base types:** `bool`, `nat`, `int`, `real`, $\dots$
- **Polymorphic type variables:** `'a`, `'b`, `'c`, $\dots$
- **Function type ($\Rightarrow$):** $\tau \Rightarrow \tau$
 - $\rightarrow f :: \text{nat} \Rightarrow \text{nat}$
 - $\rightarrow f' :: \text{nat} \Rightarrow (\text{nat} \Rightarrow \text{nat})$
- **Product type ($\times$):** $\tau \times \tau$
 - $\rightarrow f :: (\text{nat} \times \text{nat}) \Rightarrow \text{nat}$
 - $\rightarrow \text{fst} :: ('a \times 'b) \Rightarrow 'a, \quad \text{snd} :: ('a \times 'b) \Rightarrow 'b$
- **List / Set types:** $\tau \text{ list}$, $\tau \text{ set}$, user-defined.

3. Logical Properties & Theorem Statements

Specifying Logical Properties:

- Logical statements are terms of type `bool`:
 - `True :: bool`
 - `True ∧ False :: bool`
 - `[a, b] ≠ [c]`
 - `a = b → [a] = [b]`

Theorem Statements & Proof Meta-Logic:

- Natural deduction style: $A, B \vdash C$
- **Object-Logic vs Meta-Logic ($\implies$):**
 - Object-logic implication: $\longrightarrow$
 - Meta-logic implication (Isabelle meta-arrow): $\implies$
- **Lemma Formats in Isabelle:**
 - `lemma "A  $\implies$  B  $\implies$  C"`
 - `lemma "[[ A; B ]]  $\implies$  C"`
- **Top-Down Proofs:** "sorry" can be used as a temporary placeholder to "prove" a lemma while continuing top-down proof structure.